

PHYS205 Experiment 2 Lab Report: Sound Waves in a Tube

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1 Objective

In this experiment, the characteristics of sound waves in a tube are investigated. The primary goals are to explore the relationship between the tube length, wavelength, and resonant frequencies for both open and closed tubes. Additionally, the speed of sound in air is determined experimentally and compared with the theoretical value.

2 Theory

Sound Waves

When the diaphragm of a speaker vibrates, it produces a sound wave that travels through the air. This wave consists of small movements of air molecules toward and away from the speaker. If you were to observe a small volume of air near the speaker, you would notice that the air doesn't move significantly; instead, it vibrates back and forth toward and away from the speaker at the same frequency as the speaker's vibrations. This motion is quite similar to waves propagating on a string. However, there's a key difference: if you watch a small section of the string, its vibrational motion is perpendicular to the direction of wave propagation. In contrast, the motion of a small volume of air in a sound wave is parallel to the wave's direction of travel. This parallel motion is why sound waves are called *longitudinal* waves.

Another way to think about a sound wave is as a series of compressions and rarefactions. When the diaphragm of a speaker moves outward, the air near it is compressed, creating a small region of relatively high air pressure, known as a *compression*. This high-pressure region compresses the air next to it, which in turn compresses the air next to it, causing the high pressure to spread away from the speaker. Conversely, when the diaphragm moves inward, a low-pressure region of air, called a *rarefaction*, is created near the diaphragm. This rarefaction also propagates away from the speaker.

Generally, sound waves travel outward in all directions from their source. However, studying sound waves can be simplified by focusing on their motion in one dimension, as demonstrated in the Resonance Tube.

The speed of sound v is related to frequency f and wavelength λ by:

$$v = f\lambda \quad (1)$$

The theoretical speed of sound in air depends on the temperature T (in $^{\circ}\text{C}$) according to the equation:

$$v_{\text{theoretical}} = 331.5 + 0.607T \quad (2)$$

Standing Waves in a Tube

Standing waves are formed in a vibrating string when a wave reflects from one end, causing the returning wave to interfere with the original wave. Similarly, standing waves occur when a sound wave reflects from the end of a tube. A standing wave on a string has nodes points where the string remains stationary and antinodes points where the string vibrates with maximum amplitude. Analogously, a standing sound wave has displacement nodes points where the air does not vibrate and displacement antinodes points where the air's vibration amplitude is at its peak.

Pressure nodes and antinodes also exist within the waveform. In fact, pressure nodes occur at displacement antinodes, and pressure antinodes occur at displacement nodes. This can be understood by considering a pressure antinode as being situated between two displacement antinodes that vibrate 180° out of phase with each other. When the air of the two displacement antinodes moves toward each other, the pressure at the pressure antinode reaches its maximum. Conversely, when they move apart, the pressure decreases to a minimum.

Reflection of the sound wave occurs at both open and closed tube ends. If the tube end is closed, the air has no escape route, so a displacement node (a pressure antinode) must be present at a closed end. On the other hand, if the tube end is open, the pressure remains very close to room pressure, resulting in a pressure node (a displacement antinode) at an open end of the tube.

Resonance

As mentioned earlier, a standing wave forms when a wave reflects off the tube's end and the returning wave interferes with the original wave. However, the sound wave actually undergoes multiple reflections back and forth between the tube's ends, and all these reflections interfere together. Generally, the multiple reflected waves are not all in phase, resulting in a small amplitude wave pattern. However, at specific frequencies of oscillation, all the reflected waves are in phase, leading to a very high amplitude standing wave. These frequencies are known as resonant frequencies.

In this experiment, we investigate the relationship between the tube's length and the frequencies at which resonance occurs. It is demonstrated that the conditions for resonance are more easily understood in terms of the wavelength of the wave pattern, rather than the frequency.

The resonance conditions depend on the boundary conditions of the tube:

Open Tube

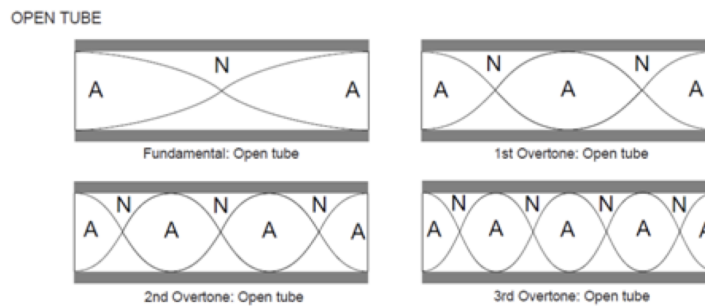


Figure 1: Resonance states in open tube

For a tube open at both ends with length of L , resonance occurs when a pressure node exists at both ends. The relationship is given by:

$$L = \frac{n\lambda}{2}, \quad n = 1, 2, 3, \dots \quad (3)$$

Taking into account the "end correction" ($0.8d$ where d is the diameter), the accurate formula is:

$$L + 0.8d = \frac{n\lambda}{2}, \quad n = 1, 2, 3, \dots \quad (4)$$

Closed Tube

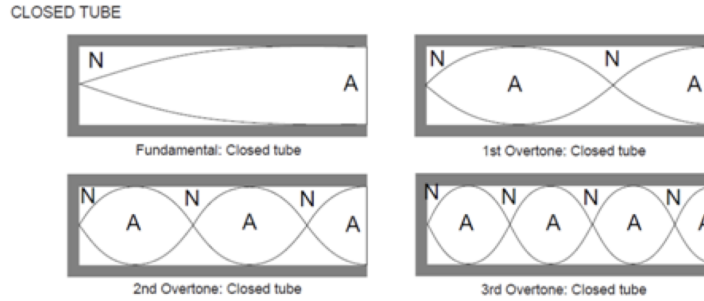


Figure 2: Resonance states in closed tube

For a tube closed at one end with length of L , resonance occurs when there is a pressure node at the open end and a pressure antinode at the closed end. The relationship is:

$$L = \frac{n\lambda}{4}, \quad n = 1, 3, 5, \dots \quad (5)$$

With the end correction ($0.4d$ where d is the diameter), the formula is:

$$L + 0.4d = \frac{n\lambda}{4}, \quad n = 1, 3, 5, \dots \quad (6)$$

3 Procedure

Equipment

- Resonance Tube

- Function Generator and Amplifier
- Oscilloscope
- Microphone with probe

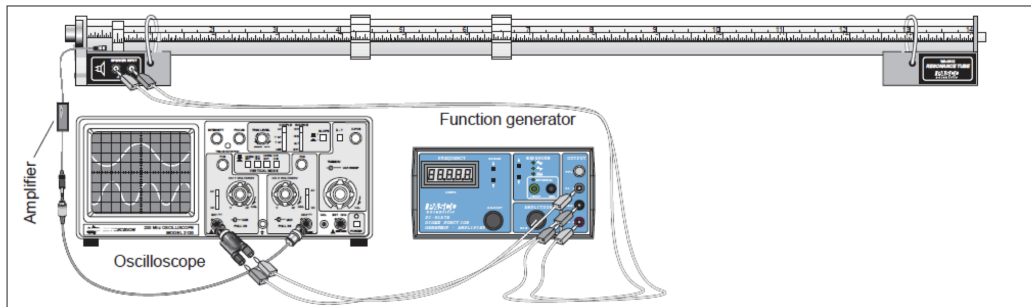


Figure 3: Equipment Setup

The experimental setup consists of a resonance tube driven by a speaker connected to a function generator as shown in Figure 3. The oscilloscope is turned on and the time/division is set to 0.2 ms/div and the gain on channel one is set to approximately 20 mV/div. Also, set the signal generator to approximately 800 Hz and gradually increase the amplitude until the speaker is clearly audible. The microphone is mounted on the end of the probe arm and insert it into the tube through the hole in the speaker/microphone stand.

Part A: Standing Waves in a Tube

A sound wave traveling down a tube reflects back and forth from its ends, causing all the waves, both the original and the reflections, to interfere with each other. When the length of the tube and the wavelength of the sound wave are such that all the waves moving in the same direction are in phase, a standing wave pattern forms. This phenomenon is known as resonance mode for the tube, and the frequencies at which resonance occurs are called resonant frequencies. In this experiment, we set up standing waves inside the Resonance Tube and use the miniature microphone to analyze the characteristics of these standing waves.

Open Tube

1. The tube is left open at one end.
2. The microphone was first located near the speaker.
3. The frequency is slowly adjusted to capture the highest amplitude.
4. Then, the frequency is recorded in Table 1.
5. As you move the microphone down the length of the tube, note the positions where the oscilloscope signal reaches its maximum and minimum values. Then, they are recorded in Table 1.

Closed Tube

1. Insert the piston into the tube, until it reaches the maximum point that the microphone can reach coming in from the speaker end.
2. The microphone was first located near the piston.
3. The frequency is slowly adjusted to capture the lowest amplitude (like a horizontal line in an oscilloscope).
4. Then, the frequency is recorded in Table 2.
5. As you move the microphone down the length of the tube, note the positions where the oscilloscope signal reaches its maximum and minimum values. Then, they are recorded in Table 1.

Part B: Tube Length and Resonant Modes

For a given tube length, there are various resonant frequencies—frequencies at which standing waves formed within the tube. Conversely, for a specific frequency, there are different tube lengths at which a standing wave was formed. In this experiment, we investigated the series of tube lengths that resonate with a particular frequency.

1. Insert the piston into the tube, until it reaches the maximum point that the microphone can reach coming in from the speaker end.

2. Set the signal generator to an arbitrary frequency near 800 Hz and turn the amplitude up until the speaker is clearly heard. Then, record this frequency.
3. Adjust the piston position carefully until you find the point which produces the largest signal on the oscilloscope screen. Then, record this position in Table 3.
4. Continue moving the piston further into the tube to find all subsequent positions which produce standing waves and record each new position.
5. Repeat the procedure for another resonant frequency (e.g., 1000 Hz) and record your results in Table 3.

4 Data

Part A: Standing Waves

Table 1: Open Tube Resonance Data

Microphone Positions (cm)	
Maxima	Minima
1	14
24	33
42	52
60	71

Resonant Frequency = 905 Hz

Table 2: Closed Tube Resonance Data

Microphone Positions (cm)	
Maxima	Minima
12.5	1
37	22
57	45
80	68

Resonant Frequency = 750 Hz

Part B: Tube Length and Resonant Modes

Table 3: Piston Positions for Resonance (Constant Frequency)

Frequency: 800 Hz	Frequency: 1000 Hz
Piston Positions (cm)	Piston Positions (cm)
74	76
52	58.5
31	41.5
9	24
—	7

5 Data Analysis

In this section, the speed of sound is calculated using the formula $v = f\lambda$. Since the distances between consecutive minima or maxima slightly different due to experimental uncertainty, the average distance (Δx_{avg}) was calculated for each trial to determine the wavelength (λ).

Part A: Standing Waves

Open Tube

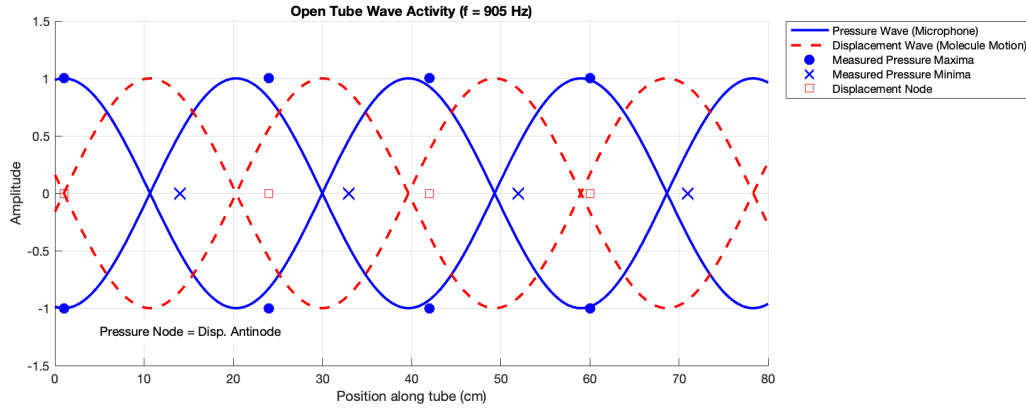


Figure 4: Wave Activity

In Figure 4, the displacement wave was calculated using the pressure wave data from the microphone and the 90° phase difference between the pressure and displacement waves. MATLAB was used for calculations and plots.

For the open tube, the distance between two consecutive maxima (or minima) is half a wavelength ($\lambda/2$). Using the data from Table 1:

- **Distance between Maxima:** $24 - 1 = \mathbf{23}$ cm, $42 - 24 = \mathbf{18}$ cm, $60 - 42 = \mathbf{18}$ cm.
- **Distance between Minima:** $33 - 14 = \mathbf{19}$ cm, $52 - 33 = \mathbf{19}$ cm, $71 - 52 = \mathbf{19}$ cm.

The average distance (Δx_{avg}) is calculated by averaging these intervals:

$$\Delta x_{avg} \approx 19.33 \text{ cm} = 0.1933 \text{ m}$$

The wavelength is:

$$\lambda_{open} = 2 \times \Delta x_{avg} = 2 \times 0.1933 = 0.3866 \text{ m}$$

Using the resonant frequency $f = 905 \text{ Hz}$:

$$v_{open} = f \lambda_{open} = 905 \times 0.3866 = 349.87 \text{ m/s} \quad (7)$$

Closed Tube

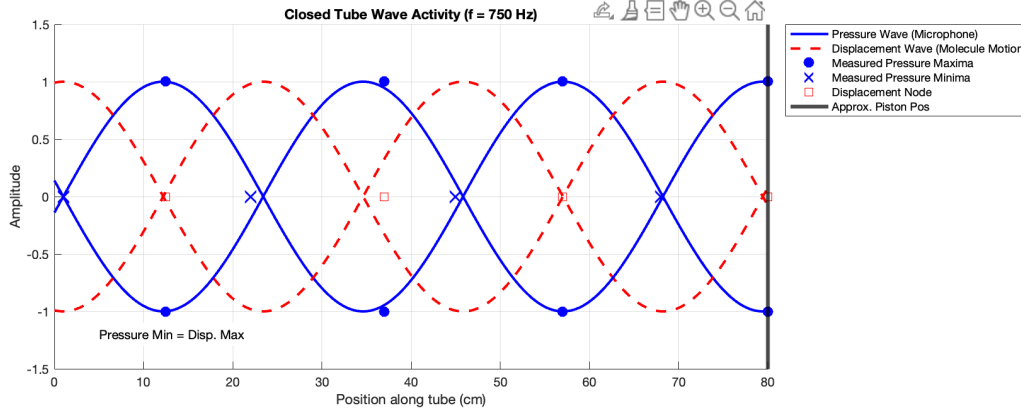


Figure 5: Wave Activity

In Figure 5, the displacement wave was calculated using the pressure wave data from the microphone and the 90° phase difference between the pressure and displacement waves. MATLAB was used for calculations and plotting.

Similar to the open tube, the distance between consecutive maxima (or minima) is $\lambda/2$. Using the data from Table 2:

- **Distance between Maxima:** $37 - 12.5 = \mathbf{24.5}$ cm, $57 - 37 = \mathbf{20}$ cm, $80 - 57 = \mathbf{23}$ cm.
- **Distance between Minima:** $22 - 1 = \mathbf{21}$ cm, $45 - 22 = \mathbf{23}$ cm, $68 - 45 = \mathbf{23}$ cm.

The average distance is:

$$\Delta x_{avg} \approx 22.42 \text{ cm} = 0.2242 \text{ m}$$

The wavelength is:

$$\lambda_{closed} = 2 \times 0.2242 = 0.4484 \text{ m}$$

Using the resonant frequency $f = 750 \text{ Hz}$:

$$v_{closed} = f\lambda_{closed} = 750 \times 0.4484 = 336.3 \text{ m/s} \quad (8)$$

Observations of Wave Nature

Based on the experimental measurements using the microphone, which is sensitive to pressure, the nature of the wave behavior at the boundaries was observed as follows:

Open Tube: We cannot reach the exactly open end of the tube because the length of the prob is 80 cm and the tube is almost 90 cm . However, the measurements at 80 cm is a pressure antinode and if we estimate the graph at 90 cm using Figure 4, the open end of the tube corresponds to a pressure node as shown in Figure 4. This satisfies our expectations because the open end is exposed to the atmospheric air pressure, which remains relatively constant. In terms of air particle motion, a pressure node corresponds to a displacement antinode. This means that at the open end, the air molecules oscillate with maximum amplitude.

Closed Tube: At the closed end of the tube, the microphone detected a pressure antinode. Because air molecules cannot move through a solid obstacle, their displacement must be zero at the piston. A point of zero displacement is a displacement node. As mentioned before, a displacement node corresponds to a pressure antinode, which explains the maximum pressure reading observed at the solid obstacle.

Part B: Variable Tube Length Analysis

In this part, the frequency was fixed, and the tube length was varied. The distance between consecutive resonance positions of the piston corresponds to $\lambda/2$.

Frequency: 800 Hz

Using the piston positions from Table 3:

$$74\text{ cm}, 52\text{ cm}, 31\text{ cm}, 9\text{ cm}$$

Distances are:

$$74 - 52 = \mathbf{22\text{ cm}}, 52 - 31 = \mathbf{21\text{ cm}}, 31 - 9 = \mathbf{22\text{ cm}}$$

The average distance is:

$$\Delta L_{avg} \approx 21.66 \text{ cm} = 0.2166 \text{ m}$$

The wavelength is:

$$\lambda = 2 \times 0.2166 = 0.4332 \text{ m}$$

Using the resonant frequency $f = 800 \text{ Hz}$:

$$v_{800\text{Hz}} = 800 \times 0.4332 = 346.56 \text{ m/s} \quad (9)$$

Frequency: 1000 Hz

Using the piston positions from Table 3:

$$76 \text{ cm}, 58.5 \text{ cm}, 41.5 \text{ cm}, 24 \text{ cm}, 7 \text{ cm}$$

Distances are:

$$76 - 58.5 = \mathbf{17.5 \text{ cm}}, 58.5 - 41.5 = \mathbf{17 \text{ cm}}, 41.5 - 24 = \mathbf{17.5 \text{ cm}}, 24 - 7 = \mathbf{17 \text{ cm}}$$

The average distance is:

$$\Delta L_{avg} \approx 17.25 \text{ cm} = 0.1725 \text{ m}$$

The wave length is:

$$\lambda = 2 \times 0.1725 = 0.345 \text{ m}$$

Using the resonant frequency $f = 1000 \text{ Hz}$:

$$v_{1000\text{Hz}} = 1000 \times 0.345 = 345.0 \text{ m/s} \quad (10)$$

Relationship to Part A Data

The analysis of Part A (Closed Tube) shows the closed end of the tube is a pressure antinode.

In Part B, we saw that resonance occurred only at specific piston positions. By analyzing the intervals, we found that the distance between these piston positions corresponds to $\lambda/2$, which is identical to the spacing between consecutive pressure maxima seen in Part A.

Error Analysis

The theoretical speed of sound depends on the room temperature (T). Assuming a room temperature of $T = 25^\circ C$:

$$v_{theo} = 331.5 + 0.607T \quad (11)$$

For $T = 25^\circ C$, $v_{theo} \approx 346.675 \text{ m/s}$.

The experimental speeds are found in previous part for experiment Part A open and closed tubes and Part B 800 Hz and 1000 Hz frequencies as:

$$v_{open} = 349.87 \text{ m/s} \quad v_{closed} = 336.3 \text{ m/s}$$

$$v_{800Hz} = 346.56 \text{ m/s} \quad v_{1000Hz} = 345 \text{ m/s}$$

respectively.

The percentage error can be calculated with this formula:

$$\%Error = \frac{|v_{theo} - v_{exp}|}{v_{theo}} \times 100 \quad (12)$$

The percentage errors of our experiment calculated using Formula 12 for v_{open} , v_{closed} , v_{800Hz} , and v_{1000Hz} are:

$$\%Error_{open} = 0.92\% \quad \%Error_{closed} = 2.99\%$$

$$\%Error_{800Hz} = 0.03\% \quad \%Error_{1000Hz} = 0.48\%$$

respectively.

The percentage error for the average speed $v_{avg} = (v_{open} + v_{closed} + v_{800Hz} + v_{1000Hz})/4 = 344.42 \text{ m/s}$:

$$\%Error_{avg} = 0.65\%$$

6 Conclusion

In this experiment, standing waves were generated in a tube to study resonance. The relationship between tube length and resonant modes was verified for both open and closed boundary conditions.

The speed of sound was experimentally determined to be $v_{avg} = 344.2 \text{ m/s}$ as a mean of all the experimental speeds. This value was compared with the theoretical value of $v_{theo} \approx 346.675 \text{ m/s}$, which shows the percentage error of 0.65%. This deviation is not a significant deviation and can be caused by experimental uncertainties, room temperature assumption, non-ideal experimental instruments, and non-precise measurements.

In conclusion, the experiment demonstrated how sound waves behave in a tube and how resonance works, also resulted very near the theoretical expectations despite some experimental errors.